Lecture 7

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

Solving $Ax = 0$, doing elimination will not change the null space

as $EAx = E0$

$(EA)\,x = 0$

↳ same vectors would be satisfying this.

(col) space might change though

$$EA \Rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

echelon form $\Rightarrow$ staircase form

number of pivots = rank of matrix

in the given example $\underline{\text{rank} = 2}$

$Ux = 0$

→ 2 pivot columns
2 free columns

→ The idea is to put any value for the free vars and then find the pivot vars

for eg in the given example x_2 / x_4 are free

then lets say $x = \begin{bmatrix} x_1 \\ 1 \\ x_3 \\ 0 \end{bmatrix}$

$$\text{So } x = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

→ So, $\boxed{\lambda \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}}$ is part of nullspace

This is not the complete nullspace, this is the nullspace when (1, 0) are chosen as free variables.

- We can find other spaces using other free vars.

eg

$$\begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 1 \end{bmatrix} \begin{matrix} 2 \\ \\ -2 \\ \\ \end{matrix}$$

$2x_3 + 4x_4 = 0$

$x_3 = -2$

$x_1 - 4 + 2 = 0$

$x_1 = 2$

$$\lambda \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

→ Another nullspace line.

→ Now in this case all the linear combinations of these 2 nullspaces we found by putting free variables as (0, 1)s is also in nullspace.

$$c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \in N(A)$$

$\equiv N(A)$ ✓

→ Now this is going to be the exhaustive nullspace for the given matrix.

→ Number of free vars determine the dimension of nullspace and their linear combination gives us the null space

$A \to m \times n$

$\text{rank}(A) \to r$

free variables = dimension of $N(A)$ = $\underline{\underline{n - r}}$

→ U → is the echelon form

we can go even further and try to make all entries above & below pivots zero.

This form is called $\underset{\downarrow}{\underline{\underline{R}}}$

reduced row echelon form

$$A \to R \to \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

→ We can even make the pivots $\underline{\underline{1}}$

$$R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

⇒ Now, lets try to solve for $\underline{Rx = 0}$

pivot cols $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ I

free cols $\begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$ F

Let $N =$ null space matrix
(matrix with spl soln cols)

So, $N = \begin{bmatrix} -F \\ I \end{bmatrix}$

in our case $\Rightarrow$ $N = \left.\begin{bmatrix} -2 & 2 \\ 0 & -2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ incomplete !!

proof $\rightarrow$ $\begin{bmatrix} I & F \end{bmatrix} \begin{bmatrix} x_{pivot} \\ x_{free} \end{bmatrix} = 0$

$I x_{pivot} + F x_{free} = 0$

$$\boxed{x_{pivot} = -F x_{free}}$$

So, we also need to switch around to make the left block I & right block F so in this case swap 2^{nd} & 3^{rd} colums.

$$N = \begin{bmatrix} -2 & 2 \\ 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix} \checkmark$$

Another matrix A^T

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{1 free col.}$$

$$F = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\boxed{\text{rank}(A) = \text{rank}(A^T)}$$

$$N = c \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$